Работа с программными пакетами

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Цели и задачи работы

Цель лабораторной работы

Получение навыков работы с репозиториями, пакетным менеджером **dnf** и утилитой **rpm** в Linux.

Процесс выполнения лабораторной работы

Работа с репозиториями

-

```
root@localhost:/home/manieyvazi# cd /etc/yum.repos.d
root@localhost:/etc/yum.repos.d# ls
epel.repo  epel-testing.repo  rocky-addons.repo  rocky-devel.repo  rocky-extras.repo  rocky.repo
root@localhost:/etc/yum.repos.d# cat epel.repo
[epel]
name=Extra Packages for Enterprise Linux $releasever - $basearch
# It is much more secure to use the metalink, but if you wish to use a local mirror
# place its address here.
#baseurl=https://download.example/pub/epel/$releasever$[releasever_minor:+z]/Everything/$basearch/
metalink=https://mirrors.fedoraproject.org/metalink?repo=epel$[releasever_minor:*-z]-$releasever&arch=$basearch
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-$releasever_major
gpgcheck=1
repo_gpgcheck=0
metadata_expire=24h
countme=1
enabled=1

[epel-debuginfo]
name=Extra Packages for Enterprise Linux $releasever - $basearch - Debug
# It is much more secure to use the metalink, but if you wish to use a local mirror
# place its address here.
#baseurl=https://download.example/pub/epel/$releasever$[releasever_minor:+z]/Everything/$basearch/debug/
metalink=https://mirrors.fedoraproject.org/metalink?repo=epel$[releasever_minor:*-z]-debug-$releasever&arch=$basearch
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-$releasever_major
gpgcheck=1
repo_gpgcheck=0
metadata_expire=24h
enabled=0

[epel-source]
name=Extra Packages for Enterprise Linux $releasever - $basearch - Source
# It is much more secure to use the metalink, but if you wish to use a local mirror
# place its address here.
#baseurl=https://download.example/pub/epel/$releasever$[releasever_minor:+z]/Everything/source/tree/
metalink=https://mirrors.fedoraproject.org/metalink?repo=epel$[releasever_minor:*-z]-source-$releasever&arch=source
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-$releasever_major
gpgcheck=1
repo_gpgcheck=0
metadata_expire=24h
enabled=0
```

Рис. 1 — Каталог /etc/yum.repos.d и файл rocky.repo

Работа с репозиториями

-.

```
root@localhost:/etc/yum.repos.d# dnf repolist
repo id                          repo name
appstream                        Rocky Linux 10 - AppStream
baseos                           Rocky Linux 10 - BaseOS
epel                             Extra Packages for Enterprise Linux 10 - x86_64
extras                           Rocky Linux 10 - Extras
root@localhost:/etc/yum.repos.d# dnf search user
Last metadata expiration check: 17 days, 16:26:01 ago on Wed 17 Dec 2025 05:57:31 AM MSK.
================================== Name & Summary Matched: user ==================================
ghc-xdg-userdirs.x86_64 : Basic implementation of XDG user directories specification
ghc-xdg-userdirs-devel.x86_64 : Haskell xdg-userdirs library development files
ghc-xdg-userdirs-doc.noarch : Haskell xdg-userdirs library documentation
ghc-xdg-userdirs-prof.x86_64 : Haskell xdg-userdirs profiling library
gnome-user-docs.noarch : GNOME User Documentation
kf6-kuserfeedback.x86_64 : Framework for collecting user feedback for apps via telemetry and
                         : surveys
kf6-kuserfeedback-console.x86_64 : Analytics and administration tool for UserFeedback servers
kf6-kuserfeedback-devel.x86_64 : Development files for kf6-kuserfeedback
kf6-kuserfeedback-doc.noarch : Developer Documentation files for kf6-kuserfeedback
kf6-kuserfeedback-html.noarch : Developer Documentation files for kf6-kuserfeedback
libuser.x86_64 : A user and group account administration library
nagios-plugins-users.x86_64 : Nagios Plugin - check_users
perl-Test-LWP-UserAgent.noarch : LWP::UserAgent suitable for simulating and testing network calls
perl-Test-LWP-UserAgent-tests.noarch : Tests for perl-Test-LWP-UserAgent
perl-User-pwent.noarch : By-name interface to Perl built-in user name resolver
python3-userpath.noarch : Cross-platform tool for adding locations to the user PATH
samba-usershares.noarch : Provides support for non-root user shares
trousers-devel.x86_64 : TrouSerS header files and documentation
trousers-lib.x86_64 : TrouSerS libtspi library
trousers-static.x86_64 : TrouSerS TCG Device Driver Library
usermode.x86_64 : Tools for certain user account management tasks
userspace-rcu.x86_64 : RCU (read-copy-update) implementation in user-space
xdg-user-dirs.x86_64 : Handles user special directories
===================================== Name Matched: user =====================================
gnome-shell-extension-user-theme.noarch : Support for custom themes in GNOME Shell
golang-github-tklauser-numcpus.x86_64 : Package numcpus provides information about the number of
                                      : CPUs in the system
golang-github-tklauser-numcpus-devel.noarch : Package numcpus provides information about the number
                                            : of CPUs in the system
golang-github-tklauser-sysconf-devel.noarch : Sysconf for Go, without using cgo
pam_ssh_user_auth.x86_64 : PAM module to help with SSH_AUTH_INFO_0
perl-LWP-UserAgent-Determined.noarch : Virtual browser that retries errors
perl-User-Identity.noarch : Maintains info about a physical person
rust-nix+user-devel.noarch : Rust friendly bindings to *nix APIs
```

Рис. 2 — Список репозиториев

Поиск пакетов

•

```
xdoinatpp-ui.noarch : GUI interface for xdoinatpp
root@localhost:/etc/yum.repos.d# dnf search nmap
================================ Name Exactly Matched: nmap ================================
nmap.x86_64 : Network exploration tool and security scanner
================================ Name & Summary Matched: nmap ==============================
nmap-ncat.x86_64 : Nmap's Netcat replacement
root@localhost:/etc/yum.repos.d# dnf info nmap
Extra Packages for Enterprise Linux 10 - x86_64                  51 kB/s |  39 kB     00:00
Extra Packages for Enterprise Linux 10 - x86_64                 1.5 MB/s | 5.6 MB     00:03
Rocky Linux 10 - BaseOS                                          11 kB/s | 4.3 kB     00:00
Rocky Linux 10 - BaseOS                                         7.1 MB/s | 6.4 MB     00:00
Rocky Linux 10 - AppStream                                       15 kB/s | 4.3 kB     00:00
Rocky Linux 10 - AppStream                                      3.9 MB/s | 2.1 MB     00:00
Rocky Linux 10 - Extras                                         8.5 kB/s | 3.1 kB     00:00
Rocky Linux 10 - Extras                                          15 kB/s | 5.4 kB     00:00
Available Packages
Name         : nmap
Epoch        : 4
Version      : 7.92
Release      : 3.el10
Architecture : x86_64
Size         : 5.6 M
Source       : nmap-7.92-3.el10.src.rpm
Repository   : appstream
Summary      : Network exploration tool and security scanner
URL          : http://nmap.org/
License      : LicenseRef-Nmap
Description  : Nmap is a utility for network exploration or security auditing.  It supports
             : ping scanning (determine which hosts are up), many port scanning techniques
             : (determine what services the hosts are offering), and TCP/IP fingerprinting
             : (remote host operating system identification). Nmap also offers flexible target
             : and port specification, decoy scanning, determination of TCP sequence
             : predictability characteristics, reverse-identd scanning, and more. In addition
             : to the classic command-line nmap executable, the Nmap suite includes a flexible
             : data transfer, redirection, and debugging tool (netcat utility ncat), a utility
             : for comparing scan results (ndiff), and a packet generation and response
             : analysis tool (nping).
```

Рис. 3 — Информация о пакете nmap

Установка пакета nmap

-.

```
root@localhost:/etc/yum.repos.d# dnf install nmap
Last metadata expiration check: 0:01:18 ago on Sat 03 Jan 2026 10:25:55 PM MSK.
Dependencies resolved.
==========================================================================================
 Package            Architecture         Version                    Repository              Size
==========================================================================================
Installing:
 nmap               x86_64               4:7.92-3.el10              appstream              5.6 M

Transaction Summary
==========================================================================================
Install  1 Package

Total download size: 5.6 M
Installed size: 24 M
Is this ok [y/N]: y
Downloading Packages:
nmap-7.92-3.el10.x86_64.rpm                                     3.2 MB/s | 5.6 MB     00:01
------------------------------------------------------------------------------------------
Total                                                           2.6 MB/s | 5.6 MB     00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                      1/1
  Installing       : nmap-4:7.92-3.el10.x86_64                                            1/1
  Running scriptlet: nmap-4:7.92-3.el10.x86_64                                            1/1

Installed:
  nmap-4:7.92-3.el10.x86_64

Complete!
root@localhost:/etc/yum.repos.d# dnf install nmap\*
Last metadata expiration check: 0:01:35 ago on Sat 03 Jan 2026 10:25:55 PM MSK.
Package nmap-4:7.92-3.el10.x86_64 is already installed.
Package nmap-ncat-4:7.92-3.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Рис. 4 — Установка пакета nmap

Удаление пакета nmap

-.

```
root@localhost:/etc/yum.repos.d# dnf remove nmap
Dependencies resolved.
================================================================================
 Package          Architecture       Version                      Repository            Size
================================================================================
Removing:
 nmap             x86_64             4:7.92-3.el10                @appstream            24 M

Transaction Summary
================================================================================
Remove  1 Package

Freed space: 24 M
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                            1/1
  Erasing          : nmap-4:7.92-3.el10.x86_64                                  1/1
  Running scriptlet: nmap-4:7.92-3.el10.x86_64                                  1/1

Removed:
  nmap-4:7.92-3.el10.x86_64

Complete!
root@localhost:/etc/yum.repos.d# dnf remove nmap\*
Dependencies resolved.
================================================================================
 Package              Architecture       Version                    Repository            Size
================================================================================
Removing:
 nmap-ncat            x86_64             4:7.92-3.el10              @AppStream           470 k

Transaction Summary
================================================================================
Remove  1 Package

Freed space: 470 k
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
```

Рис. 5 — Работа с группами пакетов

Работа с группами пакетов

-.

```
root@localhost:/etc/yum.repos.d# dnf groups list
Last metadata expiration check: 0:04:45 ago on Sat 03 Jan 2026 10:25:55 PM MSK.
Available Environment Groups:
   Server
   Minimal Install
   Workstation
   KDE Plasma Workspaces
   KDE Plasma Mobile
   Custom Operating System
   Virtualization Host
Installed Environment Groups:
   Server with GUI
Installed Groups:
   Container Management
   Development Tools
   Headless Management
Available Groups:
   Desktop accessibility
   KDE Applications
   KDE
   KDE Multimedia support
   KDE Mobile
   KDE PIM
   KDE Software Development
   KDE Frameworks 6 Software Development
   Legacy UNIX Compatibility
   Smart Card Support
   Console Internet Tools
   .NET Development
   Graphical Administration Tools
   Network Servers
   RPM Development Tools
   Scientific Support
   Security Tools
   System Tools
root@localhost:/etc/yum.repos.d# dnf groups info "RPM Development Tools"
Last metadata expiration check: 0:05:21 ago on Sat 03 Jan 2026 10:25:55 PM MSK.
Group: RPM Development Tools
 Description: Tools used for building RPMs, such as rpmbuild.
 Mandatory Packages:
   redhat-rpm-config
   rpm-build
 Default Packages:
   rpmdevtools
```

Рис. 6 — Список групп пакетов

Работа с группами пакетов

-.

```
Complete!
root@localhost:/etc/yum.repos.d# dnf history
ID     | Command line                              | Date and time    | Action(s)      | Altered
-------------------------------------------------------------------------------------------------
    12 | groupinstall RPM Development Tools        | 2026-01-03 22:32 | I, U           |   23
    11 | remove nmap*                              | 2026-01-03 22:28 | Removed        |    1
    10 | remove nmap                               | 2026-01-03 22:28 | Removed        |    1
     9 | install nmap                              | 2026-01-03 22:27 | Install        |    1
     8 |                                           | 2025-11-18 16:10 | Install        |    1
     7 |                                           | 2025-11-18 16:09 | Install        |    1
     6 | install htop                              | 2025-11-18 16:08 | Install        |    3
     5 | install fastfetch                         | 2025-11-18 16:05 | Install        |    2  <
     4 | groupinstall Development Tools            | 2025-11-18 15:59 | Install        |    1 >
     3 | update                                    | 2025-11-18 15:51 | I, U           |  244 EE
     2 | install epel-release                      | 2025-11-18 15:48 | Install        |    1 E<
     1 |                                           | 2025-11-18 14:38 | Install        | 1373 >E
```

Рис. 7 — Установка группы RPM Development Tools

Удаление группы пакетов

-.

```
root@localhost:/etc/yum.repos.d# dnf groupremove "RPM Development Tools"
Dependencies resolved.
==========================================================================================
 Package                        Architecture   Version                 Repository          Size
==========================================================================================
Removing:
 rpmdevtools                    noarch         9.6-9.el10              @appstream         198 k
Removing unused dependencies:
 python3-argcomplete            noarch         3.2.2-4.el10            @appstream         274 k
 python3-rpmautospec            noarch         0.8.1-11.el10_1         @epel              413 k
 python3-rpmautospec-core       noarch         0.1.5-2.el10_0          @epel               11 k
Removing Groups:
 RPM Development Tools

Transaction Summary
==========================================================================================
Remove  4 Packages

Freed space: 896 k
Is this ok [y/N]: yy
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                    1/1
  Erasing          : rpmdevtools-9.6-9.el10.noarch                                      1/4
  Erasing          : python3-rpmautospec-0.8.1-11.el10_1.noarch                         2/4
  Erasing          : python3-rpmautospec-core-0.1.5-2.el10_0.noarch                     3/4
  Erasing          : python3-argcomplete-3.2.2-4.el10.noarch                            4/4
  Running scriptlet: python3-argcomplete-3.2.2-4.el10.noarch                            4/4

Removed:
  python3-argcomplete-3.2.2-4.el10.noarch          python3-rpmautospec-0.8.1-11.el10_1.noarch
  python3-rpmautospec-core-0.1.5-2.el10_0.noarch   rpmdevtools-9.6-9.el10.noarch

Complete!
root@localhost:/etc/yum.repos.d# dnf history undo 6
Last metadata expiration check: 0:02:19 ago on Sat 03 Jan 2026 10:32:45 PM MSK.
Dependencies resolved.
==========================================================================================
 Package                Architecture       Version                     Repository          Size
==========================================================================================
Removing:
```

Рис. 8 — Удаление группы RPM Development Tools

История операций dnf

Проверка работы системы.

```
Complete!
root@localhost:/etc/yum.repos.d# dnf history
ID     | Command line                                | Date and time    | Action(s)      | Altered
-------------------------------------------------------------------------------------------------
    12 | groupinstall RPM Development Tools          | 2026-01-03 22:32 | I, U           |   23
    11 | remove nmap*                                | 2026-01-03 22:28 | Removed        |    1
    10 | remove nmap                                 | 2026-01-03 22:28 | Removed        |    1
     9 | install nmap                                | 2026-01-03 22:27 | Install        |    1
     8 |                                             | 2025-11-18 16:10 | Install        |    1
     7 |                                             | 2025-11-18 16:09 | Install        |    1
     6 | install htop                                | 2025-11-18 16:08 | Install        |    3
     5 | install fastfetch                           | 2025-11-18 16:05 | Install        |    2  <
     4 | groupinstall Development Tools              | 2025-11-18 15:59 | Install        |    1 >
     3 | update                                      | 2025-11-18 15:51 | I, U           |  244 EE
     2 | install epel-release                        | 2025-11-18 15:48 | Install        |    1 E<
     1 |                                             | 2025-11-18 14:38 | Install        | 1373 >E
```

Рис. 9 — История dnf и откат действия

Использование rpm (lynx)

•

```
root@localhost:/etc/yum.repos.d# dnf list lynx
Last metadata expiration check: 0:02:48 ago on Sat 03 Jan 2026 10:32:45 PM MSK.
Available Packages
lynx.x86_64                                  2.9.0-6.el10                                appstream
root@localhost:/etc/yum.repos.d# dnf install linx --downloadonly
Last metadata expiration check: 0:03:12 ago on Sat 03 Jan 2026 10:32:45 PM MSK.
No match for argument: linx
Error: Unable to find a match: linx
root@localhost:/etc/yum.repos.d# dnf install lynx --downloadonly
Last metadata expiration check: 0:03:31 ago on Sat 03 Jan 2026 10:32:45 PM MSK.
Dependencies resolved.
================================================================================
 Package            Architecture         Version                    Repository              Size
================================================================================
Installing:
 lynx               x86_64               2.9.0-6.el10               appstream               1.6 M

Transaction Summary
================================================================================
Install  1 Package

Total download size: 1.6 M
Installed size: 6.0 M
DNF will only download packages for the transaction.
Is this ok [y/N]: y
Downloading Packages:
lynx-2.9.0-6.el10.x86_64.rpm                                     4.0 MB/s | 1.6 MB     00:00
--------------------------------------------------------------------------------
Total                                                            2.1 MB/s | 1.6 MB     00:00
Complete!
The downloaded packages were saved in cache until the next successful transaction.
You can remove cached packages by executing 'dnf clean packages'.
root@localhost:/etc/yum.repos.d# find /var/cache/dnf/ -name lynx*
/var/cache/dnf/appstream-8f18e4146890fdfc/packages/lynx-2.9.0-6.el10.x86_64.rpm
root@localhost:/etc/yum.repos.d# rpm -Uhv lynx-2.9.0-6.el10.x86_64.rpm
error: open of lynx-2.9.0-6.el10.x86_64.rpm failed: No such file or directory
root@localhost:/etc/yum.repos.d# cd /var/cache/dnf
root@localhost:/var/cache/dnf# ls
appstream-8f18e4146890fdfc  baseos.solv              epel-updateinfo.solvx    last_makecache
appstream.solv              baseos-updateinfo.solvx  expired_repos.json       packages.db
appstream-updateinfo.solvx  epel-8429a44bbab5b90a    extras-0809d34265acef7d  tempfiles.json
baseos-5b8a90704d474d3e     epel.solv                extras.solv
root@localhost:/var/cache/dnf# find -name lynx
root@localhost:/var/cache/dnf# cd
root@localhost:~# cd /var/cache/dnf/
root@localhost:/var/cache/dnf# ls
```

Рис. 10 — Загрузка пакета lynx

Установка lynx

•

```
root@localhost:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -ql dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
/etc/dnsmasq.d
/usr/lib/.build-id
/usr/lib/.build-id/c9
/usr/lib/.build-id/c9/a4bb4576977dbef150b40955510f869d246b34
/usr/lib/systemd/system/dnsmasq.service
/usr/lib/sysusers.d/dnsmasq.conf
/usr/sbin/dnsmasq
/usr/share/dnsmasq
/usr/share/dnsmasq/trust-anchors.conf
/usr/share/doc/dnsmasq
/usr/share/doc/dnsmasq/CHANGELOG
/usr/share/doc/dnsmasq/DBus-interface
/usr/share/doc/dnsmasq/FAQ
/usr/share/doc/dnsmasq/doc.html
/usr/share/doc/dnsmasq/setup.html
/usr/share/licenses/dnsmasq
/usr/share/licenses/dnsmasq/COPYING
/usr/share/licenses/dnsmasq/COPYING-v3
/usr/share/man/man8/dnsmasq.8.gz
/var/lib/dnsmasq
root@localhost:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qd dnsmasq
/usr/share/doc/dnsmasq/CHANGELOG
/usr/share/doc/dnsmasq/DBus-interface
/usr/share/doc/dnsmasq/FAQ
/usr/share/doc/dnsmasq/doc.html
/usr/share/doc/dnsmasq/setup.html
/usr/share/man/man8/dnsmasq.8.gz
root@localhost:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qc dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
root@localhost:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# man dnsmasq
root@localhost:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qc dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
root@localhost:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -q --scripts dnsmasq
preinstall scriptlet (using /bin/sh):
#precreate users so that rpm can install files owned by that user

# generated from dnsmasq-systemd-sysusers.conf
getent group 'dnsmasq' >/dev/null || groupadd -r 'dnsmasq' || :
```

Рис. 11 — Установка пакета lynx

Информация о lynx

-

Рис. 12 — Файлы пакета lynx

Документация lynx

-

Рис. 13 — Документация пакета lynx

Man-страница lynx

-

Рис. 14 — Справка man по lynx

Запуск lynx

-

Рис. 15 — Работа браузера lynx

Конфигурация lynx

-

Рис. 16 — Конфигурационные файлы lynx

Выводы по проделанной работе

Вывод

В ходе работы были изучены основные приёмы управления пакетами в Linux с использованием менеджеров **dnf** и **rpm**.
Были выполнены задачи по установке, обновлению и удалению программ, исследованию содержимого пакетов, их конфигурационных файлов, документации и установочных скриптов.
Особое внимание уделялось различиям между установкой пакетов из репозиториев и установкой локальных rpm-файлов.